
Lecture 17

ChE 467/596-012

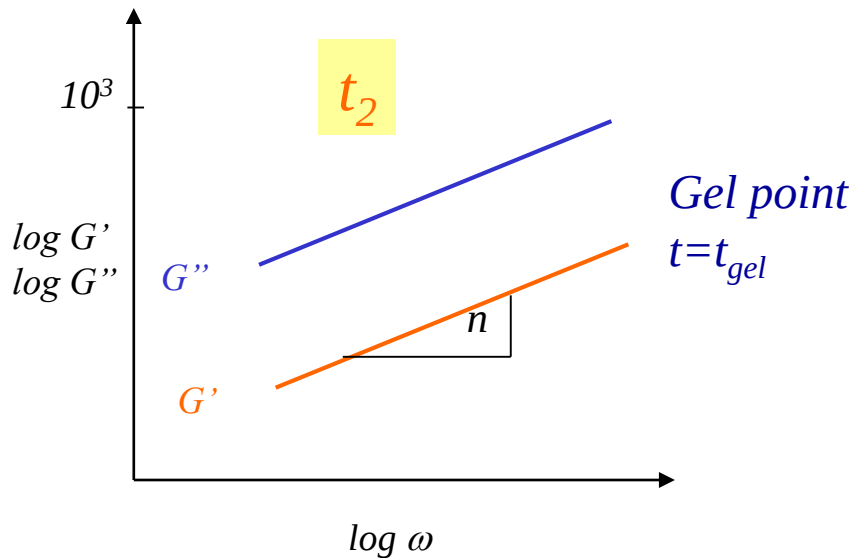
Polymer Rheology

Saad A Khan

Gel & Network-Fractals

Concept of fractals:

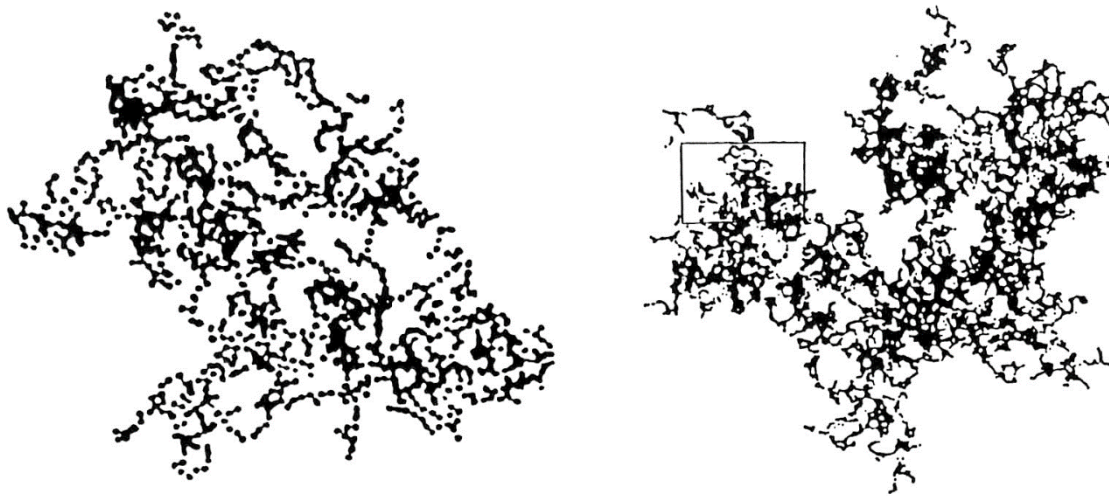
Gel point data $\Rightarrow$ *Fractal Dimension* $\Rightarrow$ *Mesh size*



n is related to fractal dimension

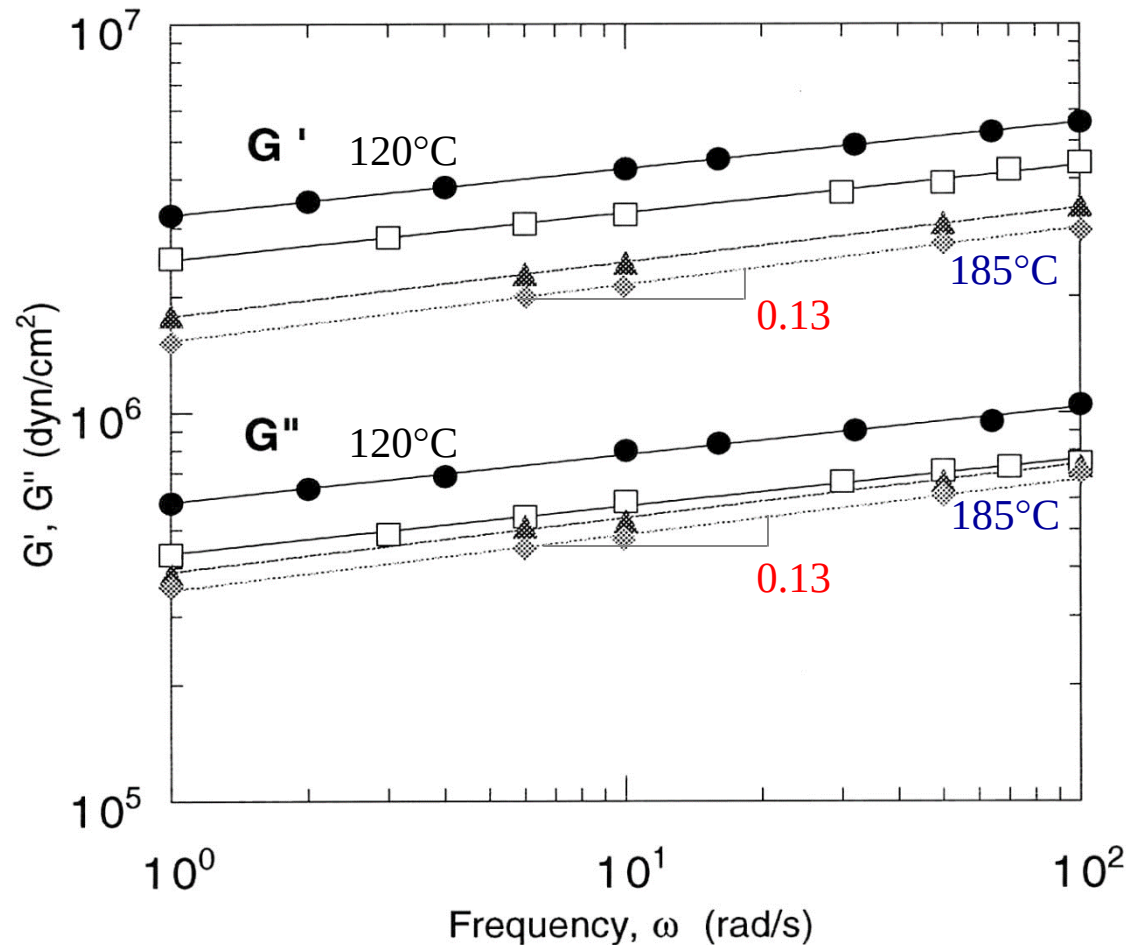
$$n = \frac{d(d+2-2d_f)}{2(d+2-d_f)}$$

Self-Similarity of Critical Gel



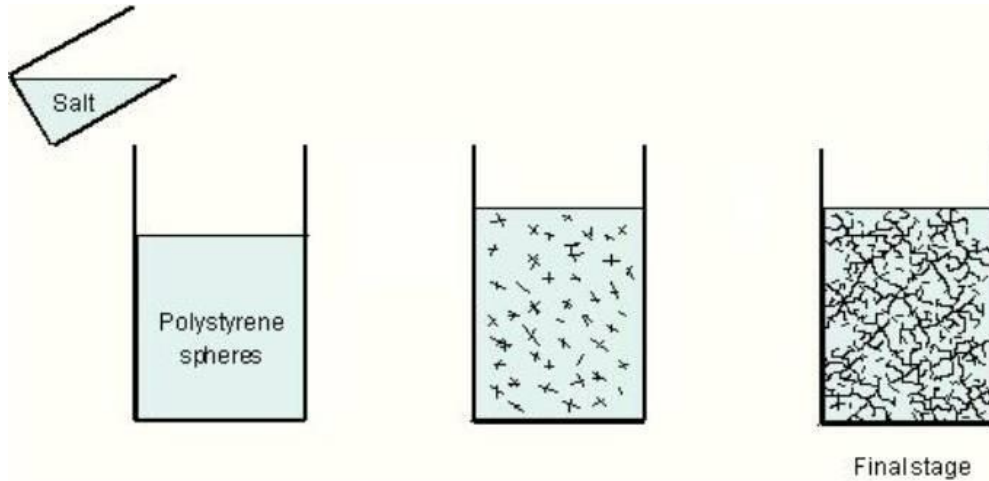
- At the gel point, the system is self-similar – a statistical fractal: $M \sim R^{d_f}$
- Fractal dimension d_f can be calculated from the value of n (Muthukumar, 1989)
- For $n = 0.13$, we get a fractal dimension $d_f = 2.38$
- High d_f indicates a dense fractal (due to high polymer MW, filler crosslinks)

Critical Gel at Different Temperatures

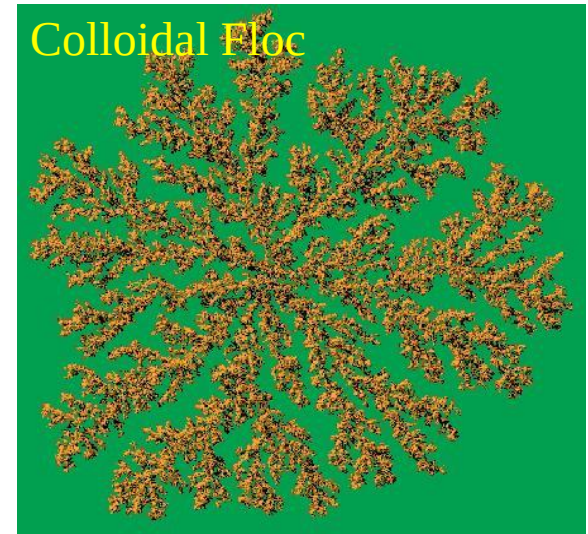


Relaxation exponent $n = 0.13$ independent of temperature

Fractal Structure of Flocs and Gels

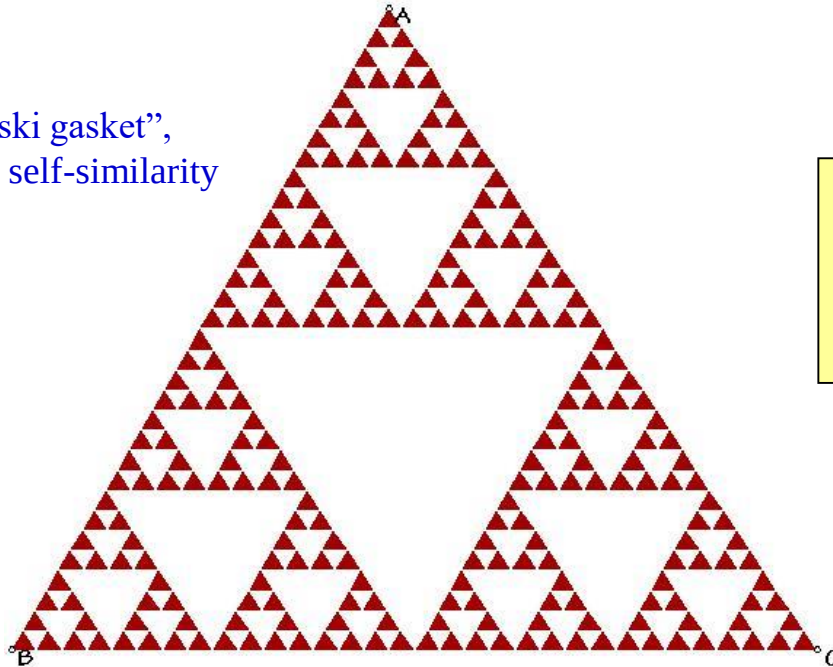


<http://www.deas.harvard.edu/projects/weitzlab/research/aging.html>



<http://www.ai.mit.edu/projects/im/broch/den1.html>

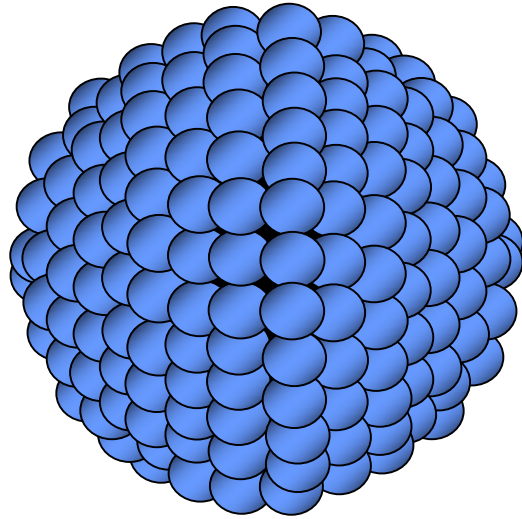
“Sierpinski gasket”,
showing self-similarity



<http://www.math.uncc.edu/~droyster/>

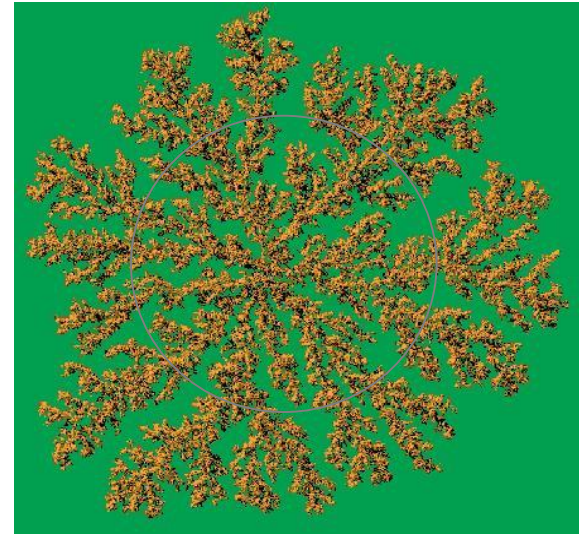
Colloidal flocs and gels tend to have a loose, open structure. They show the property of scale invariance or self-similarity and are hence referred to as fractals.

Fractal vs. Non-Fractal Clusters



Non-Fractal Spherical Cluster

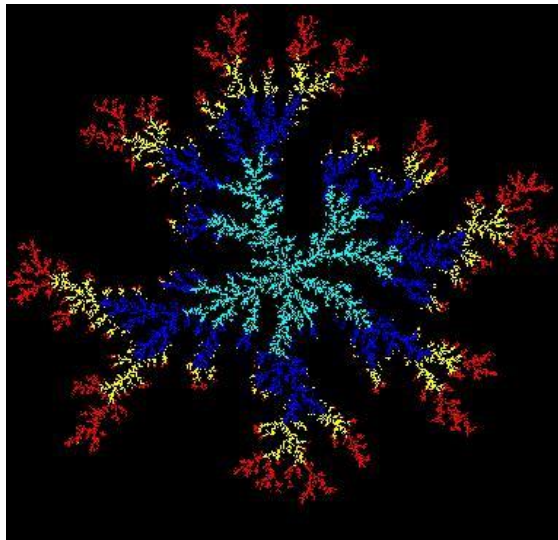
- The mass $M \sim R^3$
- Density is independent of size: $\frac{M}{V} \sim R^0$



Fractal Cluster

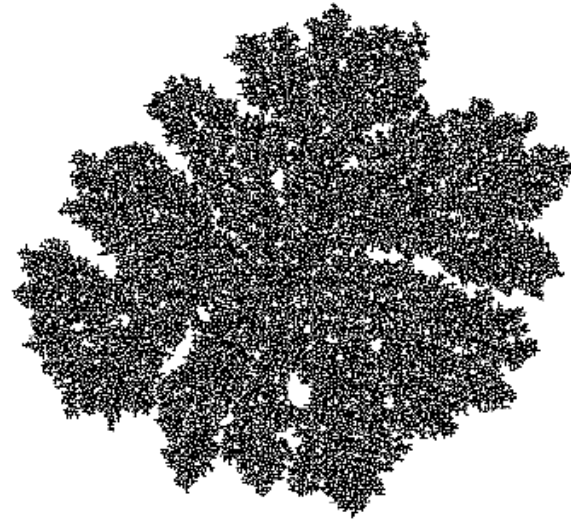
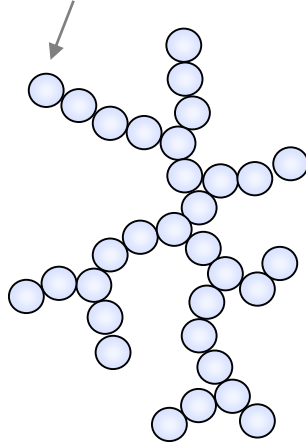
- The mass $M \sim R^{d_f}$
- Density **decreases** with size: $\frac{M}{V} \sim R^{d_f-3}$
- The fractal dimension d_f is a number < 3 .
- The lower the d_f , the **more open** the structure

Fractal Clusters and Aggregation Mechanisms



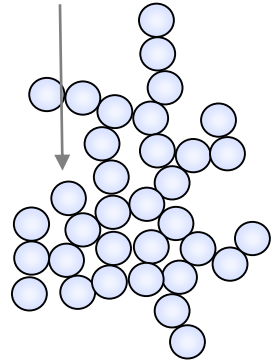
<http://www-personal.umich.edu/~lsander/>

incoming particle
sticks to the
periphery of the floc



<http://www.physics.uc.edu/~pinski/dla/dlapres/>

incoming particle
bounces around before
finding a destination



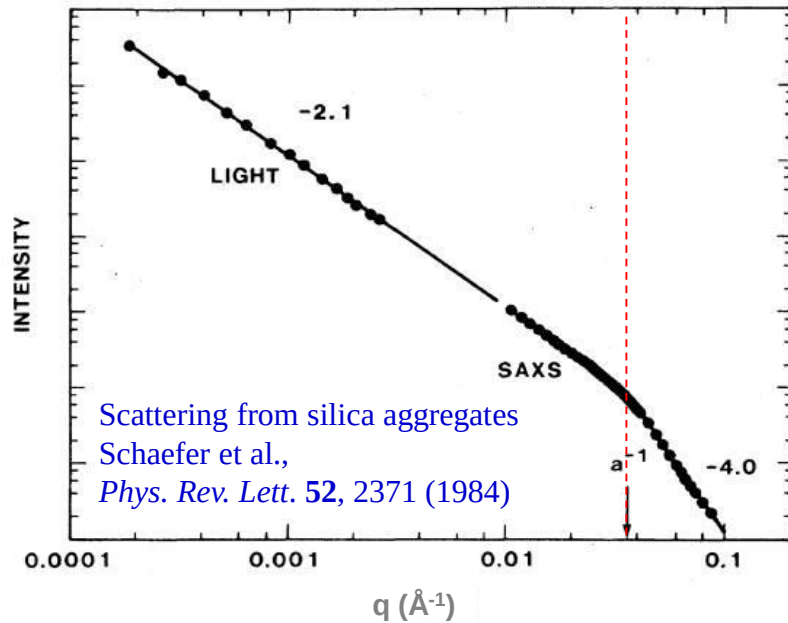
Diffusion-Limited Aggregation (DLA)

- Particles are strongly attracted to each other
- High “sticking probability”: stick on contact
- Diffusion through the fluid is the limiting step
- Leads to open flocs ($d_f = 1.75$)

Reaction-Limited Aggregation (RLA)

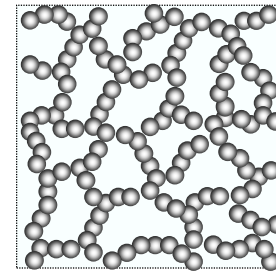
- Particles are weakly attracted to each other
- Low “sticking probability”
- Reaction is the limiting step
- Leads to more compact flocs ($d_f = 2.0 - 2.2$)

Fractal Clusters: Scattering and Rheology

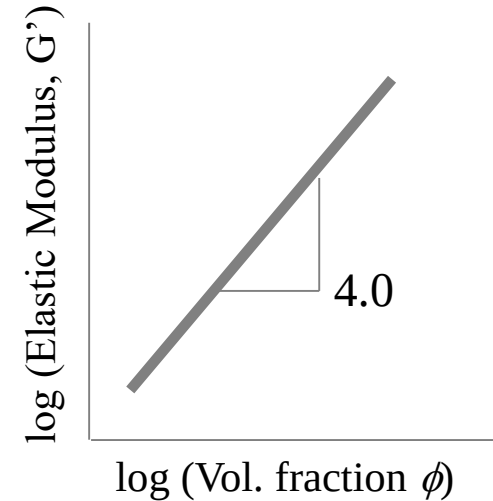


Scattering from Fractal Dispersions

- Intensity I varies as a power law: $I \sim q^{-d_f}$
- Slope gives the fractal dimension d_f
- Here $d_f = 2.1 \Rightarrow$ RLA



GEL



Rheology of Fractal Gels

- Gel $\Rightarrow$ 3-D network: G' indep. of frequency
- As part. vol. fraction $\uparrow$, $G' \uparrow$ in power law
- $G' \sim \phi^{4.0-4.5}$ for DLA ($d_f = 1.75$)

Food for thought

Loss tangent values at gel point

G' , G'' vs frequency plot shapes for gels

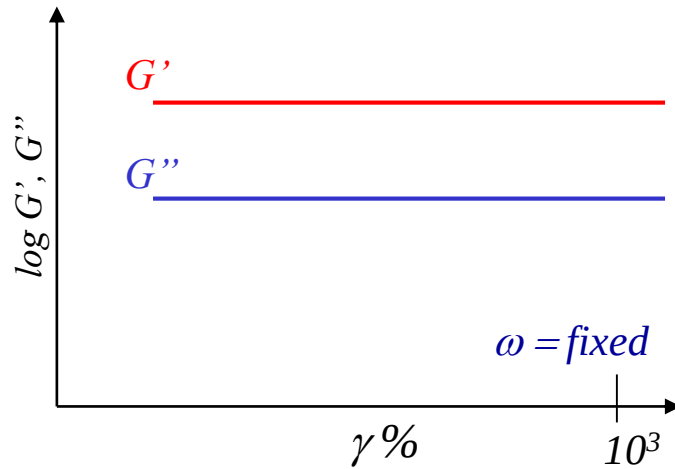
Relaxation modulus G at gel point

Implication of the flatness of G' for gels

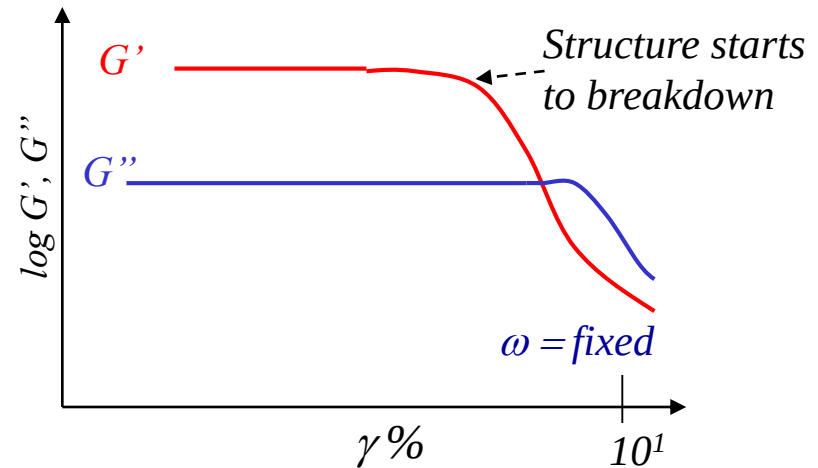
Relaxation time of a gel

Physical versus Chemical Gel Systems

Chemically α -linked gels



Physical gels

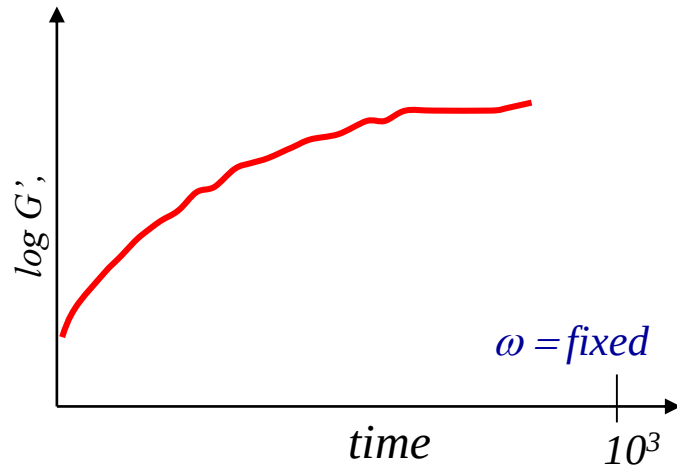


Issues

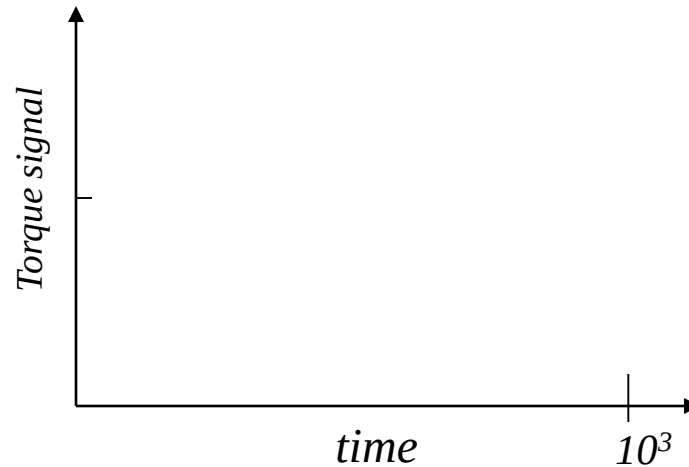
- Does the structure recover?
- To what extent is the recovery?
- What is the kinetics of recovery?
- How do we monitor recovery?
- Wall slip – why does it occur? how can we detect it?

Physical Gel Systems

Monitor recovery using dynamic experiments



- *Why dynamic experiments?*



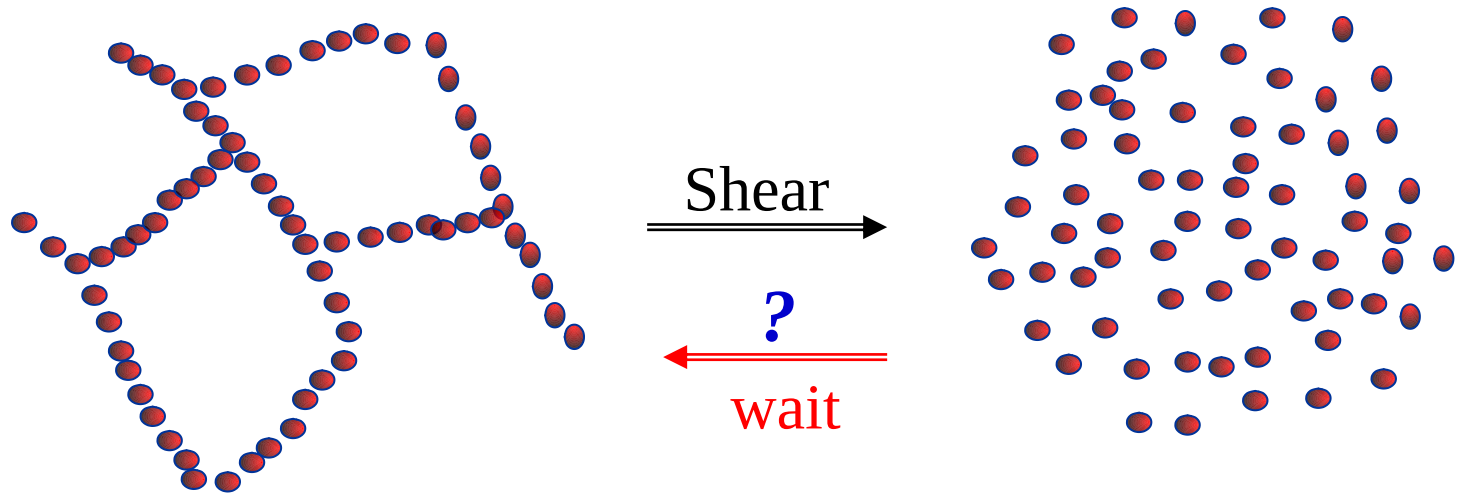
- *Plot raw torque signal*

Thixotropy

Many physical gel systems exhibit thixotropy

- Time dependent changes in rheological properties
- Not caused by chemical x-linking
- Occurs due to microstructural changes within sample
- Mostly reversible behavior
- Examples:
 - Coating/paints
 - Slurries
 - Colloidal systems

Thixotropic Material-example



- H-Bonding
- Colloidal forces
(van der Waals, electrostatic)

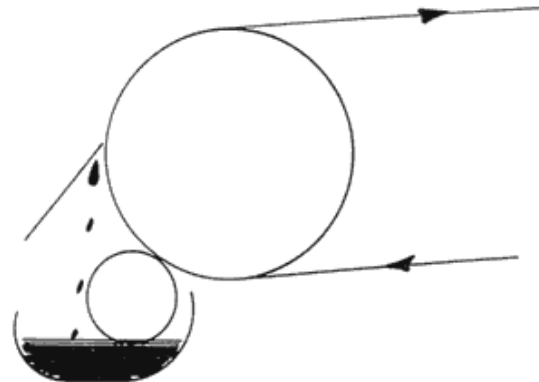
Thixotropic Material

Real Example – Paint

Is there any benefit to having thixotropy?

- brushing should break structure
- after brushing, want structure to build up fast

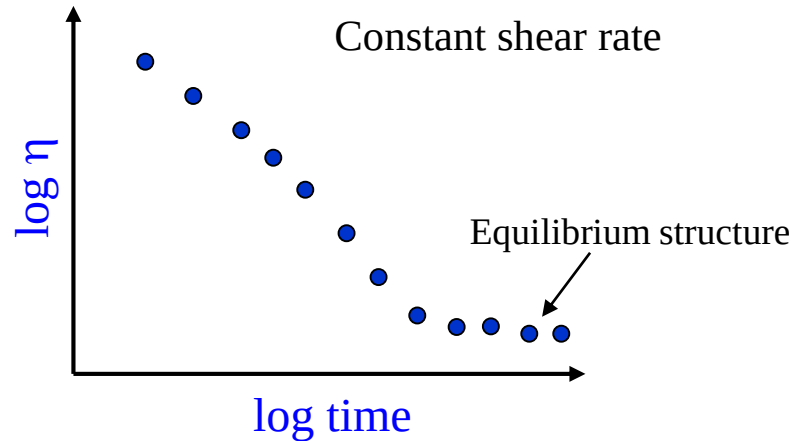
– Plastisols of PVC (Synthetic leather)



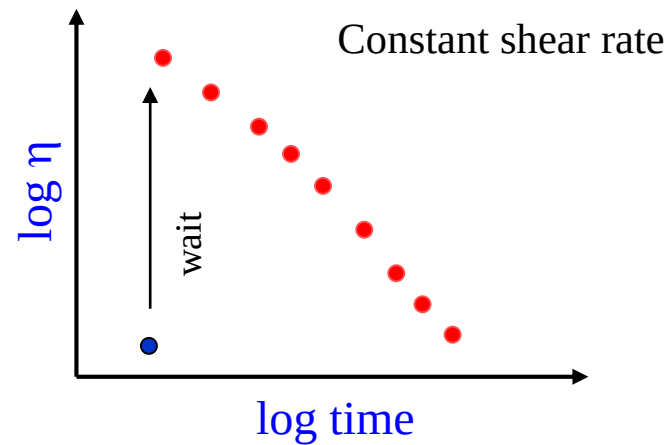
Thixotropy, cont'd

Example:

Equilibrium structure
reached after some time



Wait for $t=t'$



Thixotropy-Critical Issues

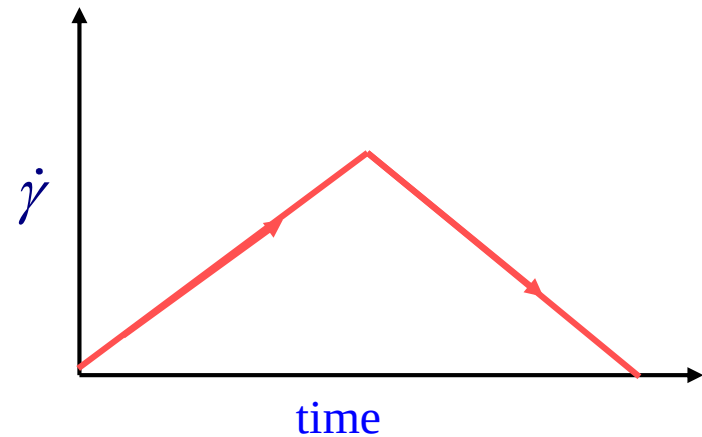
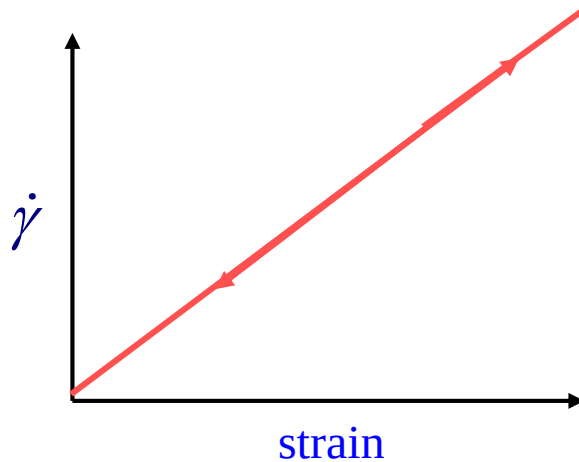
- Need to determine extent of structure breakdown
- Need to determine extent of structure recovery
- Need to get kinetics of breakdown or recovery
- How do we get reproducible data?

Thixotropy-Hysteresis

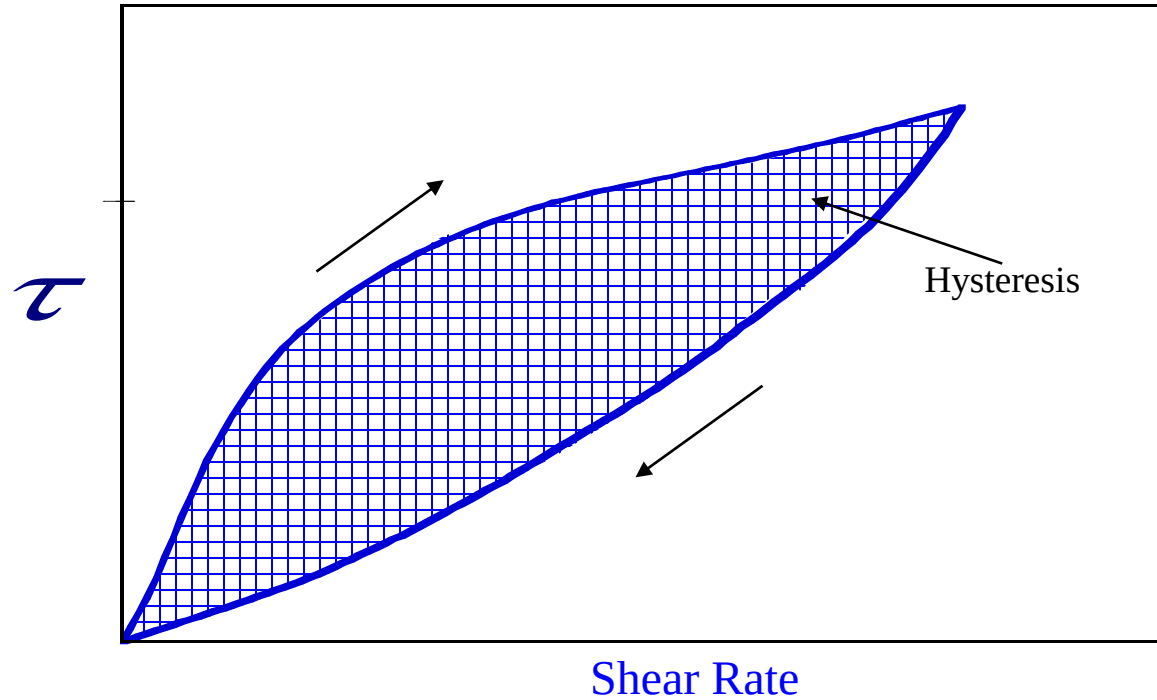
Most common experiment done to measure thixotropy

Hysteresis / thixotropic loop

Jack up shear rate at a
certain rate (0-100 s⁻¹ in min)

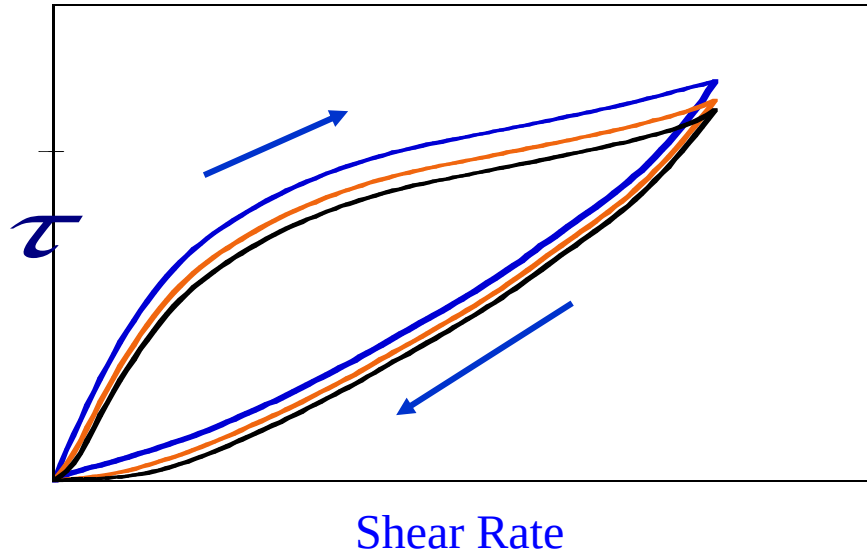


Thixotropy-Hysteresis, cont'd



Larger area => more thixotropic

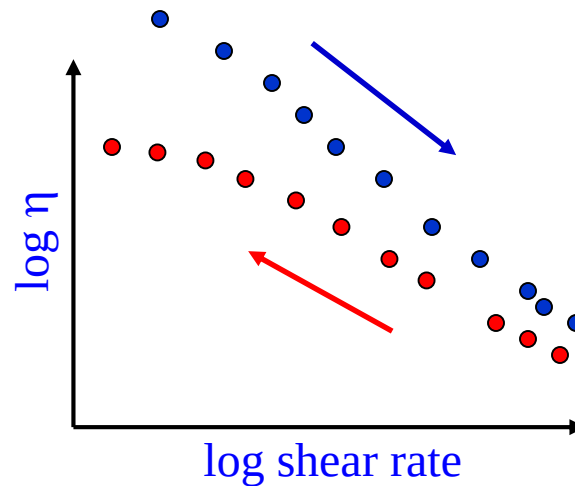
Thixotropy-Hysteresis, cont'd



Problems:

- Even at initial point some structures may have already been broken
- Several loops $\Rightarrow$ thixotropic nature decreases
- No steady state reached at a particular shear rate

Better approach to examining hysteresis?



Is this a better approach?

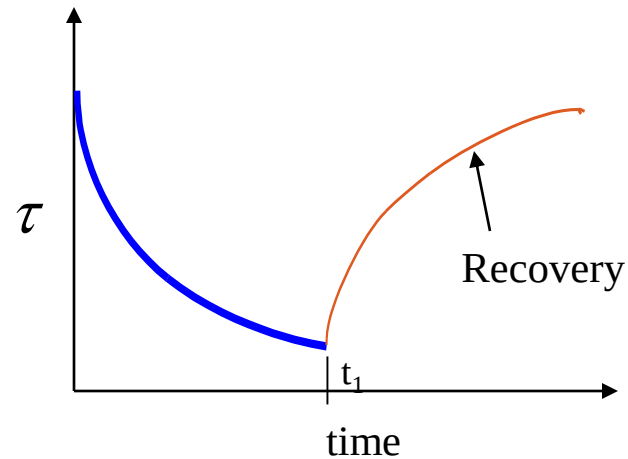
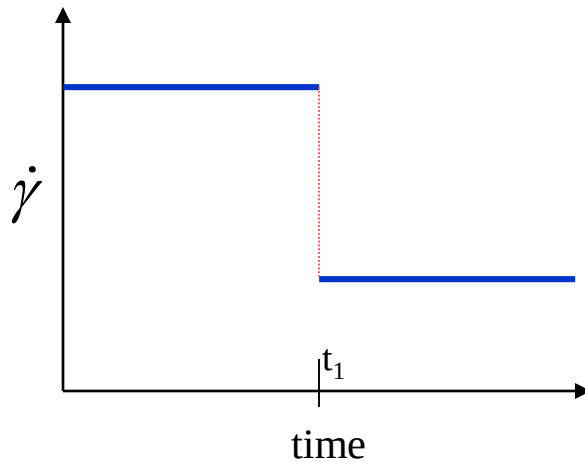
- Steady state reached for each shear rate

Can you explain the shapes of the curves? No zero shear plateau starting out but Newtonian region on the return rate sweep

Thixotropic Materials-cont'd

How do we measure recovery of structure?

Can do steady shear



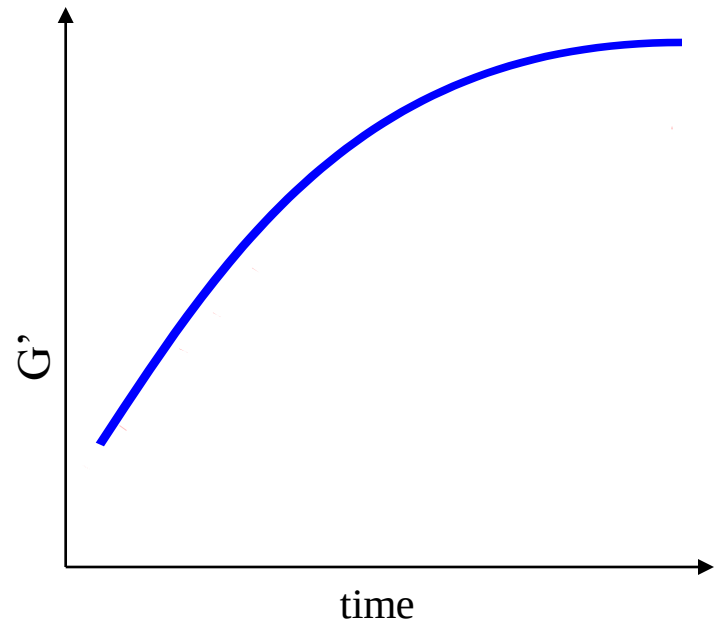
High shear rate will break all structures

Extent of recovery may be affected by the shearing, even at the low shear rate

Thixotropy-cont'd

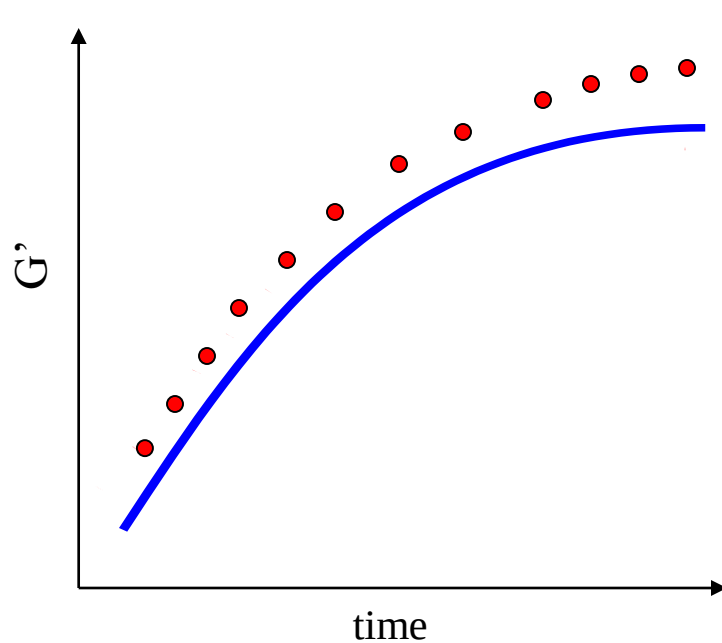
Measuring recovery of structure following shear:

- Follow recovery with dynamic experiment
- Will oscillations affect build-up?



Thixotropy-cont'd

Will oscillations affect build-up/structure recovery?

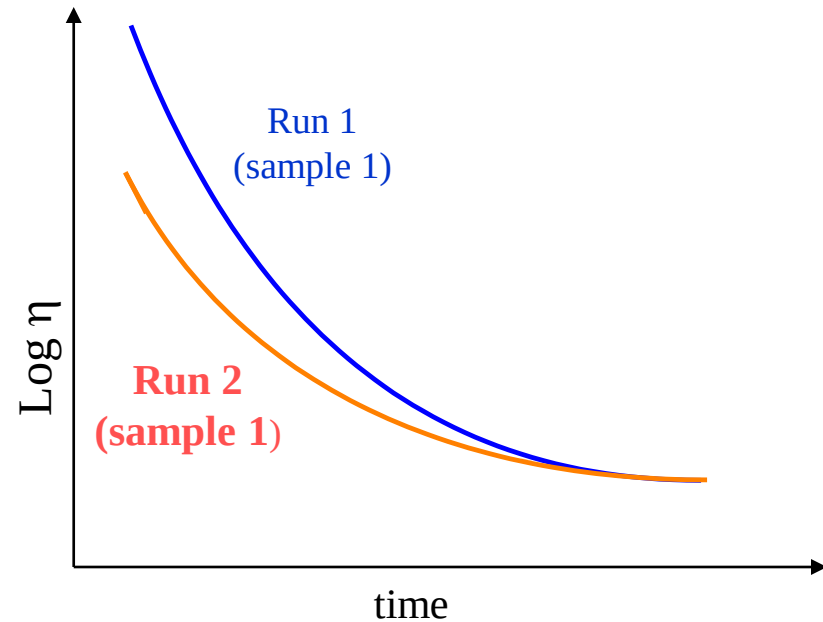
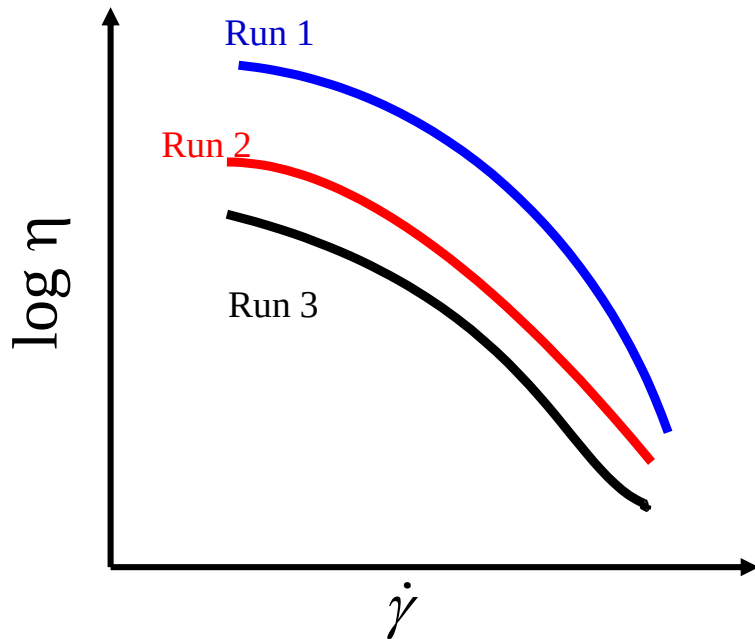


If this is the case then
oscillations are affecting
build-up

- Do continuous *versus* intermittent oscillations

Thixotropy & Reproducibility

Problem with getting reproducible data:



While loading the sample, you break or rearrange microstructure

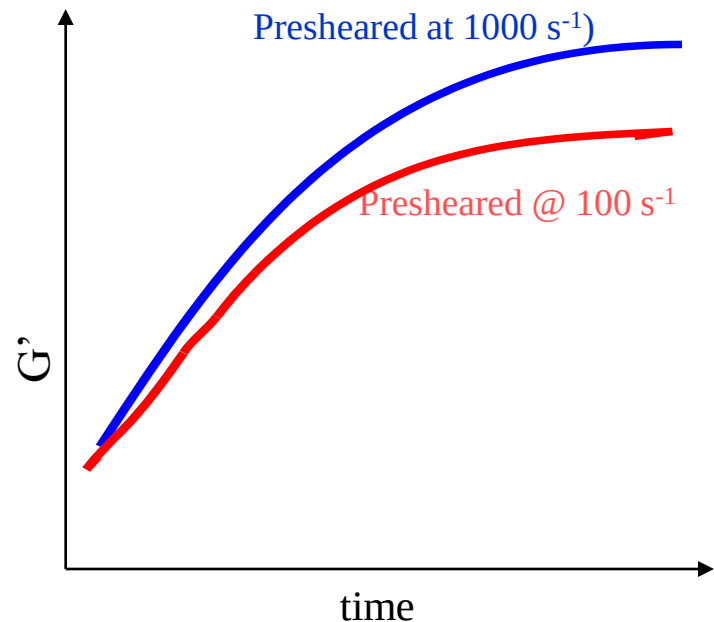
Thixotropy & Reproducibility

Need consistent initial conditions:

- Load sample slowly
- Wait a specified period of time (e.g., 15 min) before starting experiment

- Pre-shear material at specified shear rate (high)
- Wait a fixed time before starting experiment

- Shear the “heck” out
- Follow recovery with dynamic experiment

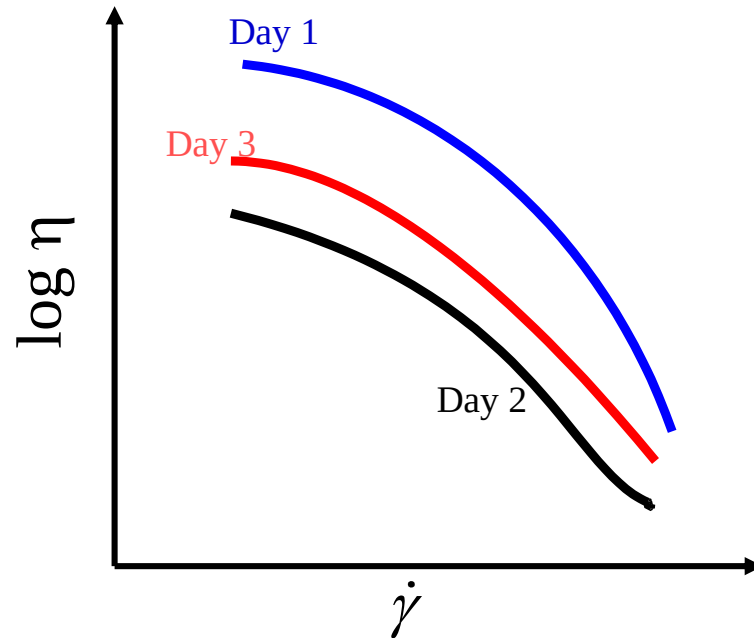


Reproducibility-cont'd

Reasons for data differences

- Chemical change/Contaminants
- Calibration not correct
- Temperature/humidity
- Evaporation
- Mixing of sample
- Thixotropy => different initial conditions

Runs for the same sample:

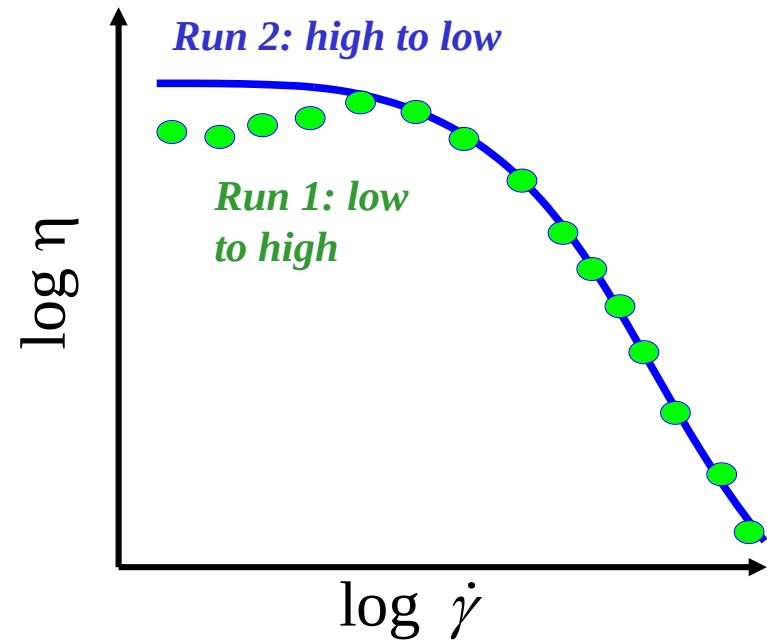


Reproducibility-cont'd

- Run 1 conducted immediately after preshear
- Run 2 conducted immediately after Run 1

Sample is thixotropic and structure needs more time to reheal

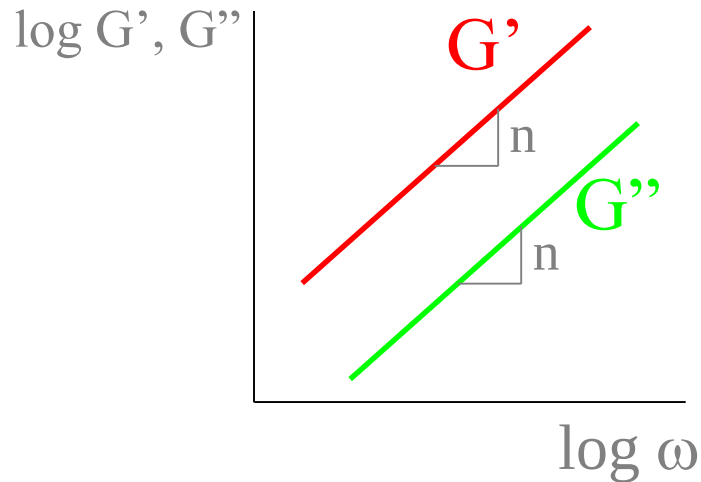
- Wait enough/longer after pre-shear



Criterion for Gel Point

Winter-Chambon Criterion

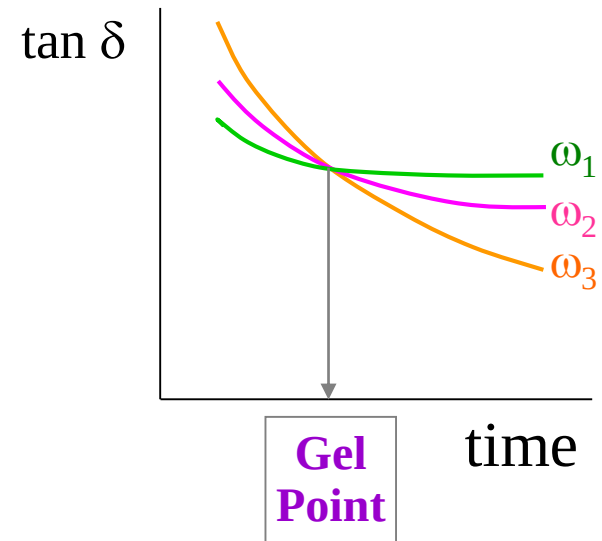
Frequency sweeps at gel point



Elastic Modulus $G' \sim \omega^n$

Viscous Modulus $G'' \sim \omega^n$

Loss tangents at different frequencies

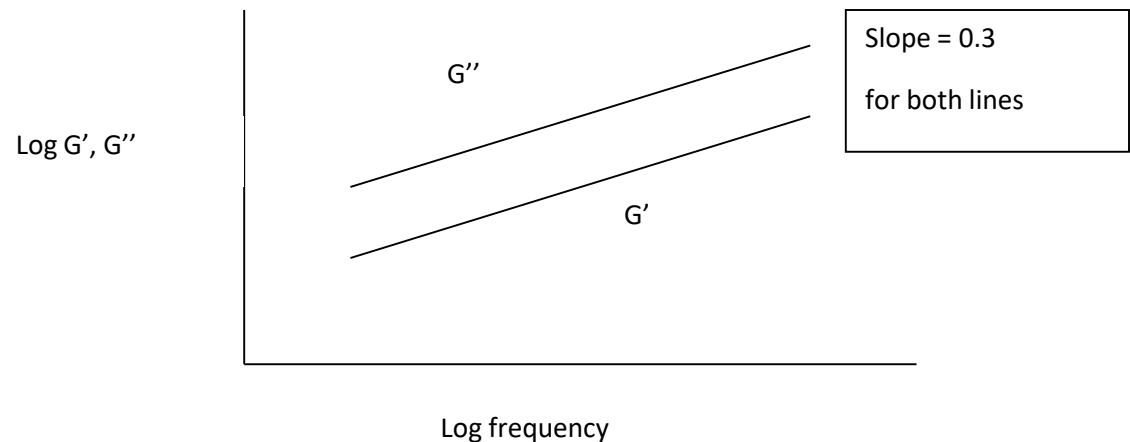


$$\tan \delta = \frac{G''}{G'} = \tan \left(\frac{n\pi}{2} \right)$$

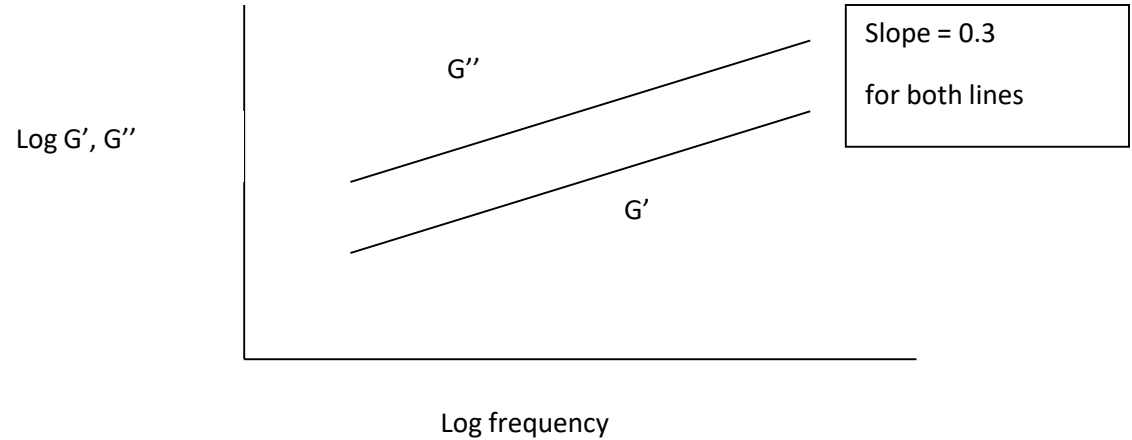
(**independent** of frequency)

In-class problem

Many students from the Polymer Morphology Group end up doing research in rheology (because this is an exciting area with more job opportunities!). During gelation of a system, one such student walked into Prof. R.J. Morphology's office rather excited. He/She showed the following data of G' and G'' versus frequency (ω) and claimed that finally he/she has found the gel point. Comment on the data. A very open-ended question!! You can talk about yes it is the gel point and why, it is not the gel point and why not, his experiments have been impeccable, etc. etc.



In-class problem



Class ended here!

In-class problem

Your synthetic chemist friend is crosslinking polymer A. At some point during the reaction, the material has passed the gel point. G' vs frequency data shows a nice flat line. Now, the theory of rubber elasticity says that $G' = nkT$ (n =number of crosslinks, k =Boltzmann constant and T =temperature in deg. K). To test this hypothesis, you measured G' of this sample at T_1 . You then heated the sample to T_2 where $T_2 = 2T_1$ (all temps are in deg. K). However, you found that G' at T_2 is more than twice the G' at T_1 . Explain your answer.

In-class problem

A photocrosslinkable system is being crosslinked in the rheometer by shining UV light. You are interested in monitoring the gelation and obtaining the gel point by running frequency sweeps at different time intervals. The problem you have is that the sample gets fully crosslinked in 20 mins, and running a reasonable frequency sweep takes 10 minutes. Are there any options available for you to monitor this crosslinking and obtain gel point?